A–B. Counting Rules, Arrangements, and Selections

Sum and product rules

Use the **sum rule** for disjoint alternatives: $m + n$ ways to do one of two tasks. Use the **product rule** for a sequence of choices: mn ways when every first choice is followed by n second choices. If alternatives overlap, use inclusion–exclusion:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

A.1: disjoint alternatives

Problem. A cafe offers 5 soups and 4 salads. A lunch includes either one soup or one salad, but not both. How many lunches are possible? Name the counting rule.
Worked solution. The soup lunches and salad lunches are disjoint alternatives. By the sum rule, add their counts: $5 + 4$.
Answer. 9 lunches, by the sum rule.

A.2: successive positions

Problem. A password contains two uppercase English letters followed by three decimal digits. Repetition is allowed. How many passwords are possible? Give an unsimplified product and name the counting rule.
Worked solution. There are 26 choices for each letter and 10 for each digit. Each choice can follow every earlier choice, so the product rule gives $26 \cdot 26 \cdot 10 \cdot 10 \cdot 10$.
Answer. $26^2 10^3$ passwords, by the product rule.

A.3: functions

Problem. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, \ldots, 8\}$. How many functions $f : A \to B$ exist? Identify the sequence of choices being counted.
Worked solution. For each of the five labeled inputs, choose any one of the eight images in B. Images may repeat: an arbitrary function need not be injective. Multiply the five independent choices.
Answer. 8^5 functions; the five choices specify $f(1), \ldots, f(5)$.

A.4: overlap

Problem. If finite sets satisfy $|A| = 19$, $|B| = 20$, and $|A \cap B| = 8$, find $|A \cup B|$. Write the formula and name the principle.
Worked solution. Adding $|A| + |B|$ counts every element of $A \cap B$ twice. Inclusion–exclusion removes one copy: $|A \cup B| = |A| + |B| - |A \cap B| = 19 + 20 - 8$.
Answer. $|A \cup B| = 31$, by inclusion–exclusion.

Generalized pigeonhole principle

Placing N objects into k boxes guarantees some box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects.
Problem (A.5). A hash table maps 40 distinct keys to 20 possible hash values. What bucket occupancy is guaranteed? State the generalized pigeonhole calculation and explain why no larger occupancy is guaranteed.
Worked solution. Treat keys as objects and hash values as boxes. The guaranteed occupancy is $\lceil 40/20 \rceil = 2$. Two keys per bucket is a possible distribution, so occupancy 3 is not forced.
Answer. At least one bucket contains at least 2 keys; no larger bound is guaranteed.

Permutations and combinations

When order matters, select and arrange:

$$P(n, r) = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}.$$

When order does not matter, select:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

For repeated object types with multiplicities $n_1, \ldots, n_t$ totaling n, the number of distinct arrangements is $n!/(n_1! \cdots n_t!)$.

B.1: committees

Problem. A committee of 3 students is selected from 10 students. How many committees are possible? Give a binomial-coefficient expression and explain why order does not matter.
Worked solution. A committee is a three-element subset. Listing its members in a different order does not create another committee. Thus choose three of ten without assigning roles.
Answer. $\binom{10}{3}$ committees.

B.2: labeled offices

Problem. A president and a vice president are selected from 10 students, with no student holding both offices. How many outcomes are possible? Give a permutation or product expression and explain why order matters.
Worked solution. Choose the president in 10 ways, then a different vice president in 9 ways. Exchanging their offices changes the outcome, so the roles are ordered.
Answer. $P(10, 2) = 10 \cdot 9 = 90$ outcomes.

B.3: repeated letters

Problem. How many distinct strings can be made by reordering the letters of `BANANA`? Give the appropriate multinomial expression.
Worked solution. First pretend the six letters are distinct, giving 6! orders. The three A's can be permuted in 3! ways and the two N's in 2! ways without changing the visible string. Divide by these duplicate counts.
Answer. $6!/(3!2!) = 60$ strings.

B.4: choosing positions

Problem. How many bit strings of length 8 contain exactly three 1s? Give a binomial-coefficient expression and identify what is being selected.
Worked solution. Choose the three positions occupied by 1s. Every remaining position is forced to be 0, and each position subset determines exactly one string.
Answer. $\binom{8}{3} = 56$ strings.

Binomial theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

Problem (B.5). Find the coefficient of x^3 in $(1+x)^8$. State the binomial-theorem term that determines your answer.
Worked solution. The term with $k = 3$ is $\binom{8}{3}1^5x^3$. Equivalently, choose the three factors that contribute x; all other factors contribute 1.
Answer. The coefficient is $\binom{8}{3} = 56$.

Frequent A–B errors

Do not add sequential choices or multiply overlapping alternatives. Decide whether changing order creates a new outcome before choosing $P(n,r)$ or $\binom{n}{r}$. Divide only for genuinely indistinguishable repetitions.

A–B readiness

Can I name the rule, show an unsimplified expression, explain order, and state what each factor or selected position represents?

C. Constrained Counting

Four reliable strategies

1. **Build directly**: multiply successive legal choices.
2. **Split into disjoint cases**: count each case, then add.
3. **Use a complement**: total minus the forbidden cases.
4. **Choose positions or members**: use binomial coefficients, then fill remaining roles.

State the sample objects and why the cases neither overlap nor leave gaps.

C.1: exact composition

Problem. How many five-card hands from a standard 52-card deck contain exactly two hearts? A hand is an unordered selection without replacement; the deck has 13 hearts and 39 nonhearts. Give an unsimplified product of binomial coefficients.
Worked solution. 1. Translate "exactly": a five-card hand with exactly two hearts has two hearts and three nonhearts. **2. Choose within each category:**

$$\binom{13}{2}\binom{39}{3}.$$

3. Justify the product: each qualifying hand has one unique pair of hearts and triple of nonhearts, so multiply the two unordered selections.
Answer. $\binom{13}{2}\binom{39}{3}$ hands.

C.2: count by complement

Problem. How many bit strings of length 8 contain at least two 1s? Use a complement and show the subtracted cases.
Worked solution. 1. Count all: there are 2^8 eight-bit strings. **2. Identify the complement:** "at least two 1s" fails only for zero or one 1. **3. Subtract the disjoint forbidden cases:**

$$2^8 - \binom{8}{0} - \binom{8}{1}.$$

The complement is shorter because it has only these two cases.
Answer. $2^8 - \binom{8}{0} - \binom{8}{1} = 247$ strings.

C.3: mixed committee

Problem. A group of 5 students is selected from disjoint pools of 12 computer-science majors and 8 mathematics majors. How many groups contain exactly 3 computer-science majors? Give an unsimplified product of binomial coefficients.
Worked solution. A five-person group uses disjoint pools of twelve CS majors and eight math majors. Exactly three CS majors therefore requires two math majors. Choose the two unordered subsets in

$$\binom{12}{3}\binom{8}{2}$$

ways. "Exactly" fixes both category counts.
Answer. $\binom{12}{3}\binom{8}{2}$ groups.

C.4: injective functions

Problem. Let A, B be finite sets with $|A| = 5$ and $|B| = 8$. How many injective functions $f : A \to B$ exist? Give a permutation or product expression and explain the restriction.
Worked solution. For $f : A \to B$ with $|A| = 5$ and $|B| = 8$, **1. label the five inputs. 2. Enforce injectivity:** after an output is used, it is unavailable. The successive choices are

$$8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = P(8, 5).$$

3. Check overcounting: input labels already distinguish the positions, so do not divide by 5!.
Answer. $P(8, 5) = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4$ injective functions.

C.5: codes with a first-position rule

Problem. A six-character code uses uppercase English letters without repetition and must begin with a vowel. Treat A, E, I, O, U as the five vowels. How many codes are possible? Give an unsimplified product.
Worked solution. 1. Enforce the special position first: a six-letter code beginning with a vowel has five first-character choices. **2. Remove that letter:** without repetition, the remaining ordered choices are $25, 24, 23, 22, 21$:

$$5 \cdot 25 \cdot 24 \cdot 23 \cdot 22 \cdot 21.$$

Each factor now has a visible job: the first enforces the vowel rule, and the others enforce no repetition.
Answer. $5 \cdot 25 \cdot 24 \cdot 23 \cdot 22 \cdot 21$ codes.

Avoiding overcounting

A counting method is valid only if each desired object is generated exactly once. Ask:

- Can the same outcome arise from two different choices or cases?
- Did I assign an order to an unordered object?
- Did I forget labels that make positions distinct?
- Are my cases exhaustive and disjoint?

Frequent C errors

"At least" usually needs several cases or a complement. "Without replacement" changes later factors. A restriction on the first character must be enforced before counting unrestricted remaining positions. Never use n^r for an injective function unless $r \leq 1$.

C readiness

Can I construct the count two ways, identify possible overlap, justify every factor, and leave a correct unsimplified expression?

D. Probability Models

Sample spaces and events

A **sample space** Ω lists all possible outcomes; an **event** is a subset of Ω. For equally likely outcomes,

$$P(A) = \frac{|A|}{|\Omega|}.$$

Always justify equal likelihood. Useful identities are

$$P(A^c) = 1 - P(A),$$
$$P(A \cup B) = P(A) + P(B) - P(A \cap B),$$
$$P(A \cap B) = P(A)P(B \mid A).$$

D.1: coin and die

Problem. A fair coin is tossed independently of a fair six-sided die roll. Find the probability that the coin is heads or the die is even (or both). Show the favorable-outcome count or an equivalent probability calculation.
Worked solution. Model: toss a fair coin and roll a fair die independently. **Sample space:** there are $2 \cdot 6 = 12$ equally likely coin–die pairs. **Favorable count:** heads gives 6, an even die gives 6, and their intersection gives 3. Hence $6 + 6 - 3 = 9$ favorable outcomes and $P = 9/12 = 3/4$. The subtraction removes the duplicate intersection.
Answer. $3/4$.

D.2: probability identities

Problem. Events A, B satisfy $P(A) = 0.55$, $P(B) = 0.40$, and $P(A \cap B) = 0.25$. Find $P(A \cup B)$ and $P(A^c \cap B)$, showing the identities used.
Worked solution. If $P(A) = 0.55$, $P(B) = 0.40$, and $P(A \cap B) = 0.25$, then

$$P(A \cup B) = 0.55 + 0.40 - 0.25 = 0.70.$$

Also $B = (A \cap B) \dot{\cup} (A^c \cap B)$, so $P(A^c \cap B) = 0.40 - 0.25 = 0.15$.
Answer. $P(A \cup B) = 0.70$ and $P(A^c \cap B) = 0.15$.

D.3: uniform bit strings

Problem. An 8-bit string is chosen uniformly from all 2^8 such strings. Find the probability that it contains exactly three 1s. Give an exact fraction.
Worked solution. Denominator: 2^8 equally likely eight-bit strings. **Numerator:** choose the three positions containing 1s, so $\binom{8}{3}$ strings qualify. Thus

$$P(\text{exactly three 1s}) = \frac{\binom{8}{3}}{2^8}.$$

The numerator and denominator must describe the same sample space.
Answer. $\binom{8}{3}/2^8 = 7/32$.

Conditional probability and independence

For $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

A and B are independent exactly when $P(A \cap B) = P(A)P(B)$, equivalently $P(A \mid B) = P(A)$ when defined. Independence is a claim to test, not a default assumption.

D.4: without replacement

Problem. A bag contains 4 red and 6 blue tokens. Two tokens are drawn uniformly, one at a time, without replacement. Find the probability that the first is red and the second is blue. Show the conditional product.
Worked solution. A bag has four red and six blue tokens. **First draw:** red has probability 4/10. **Update:** after removing a red, nine tokens remain and all six blue tokens remain. Therefore

$$\frac{4}{10} \cdot \frac{6}{9}.$$

The second factor is conditional on the first draw; keeping denominator 10 would describe replacement, not this experiment.
Answer. $(4/10)(6/9) = 4/15$.

D.5: two different conditionals

Problem. In a class, 24 students use Python, 18 use Java, and 10 use both. One student is chosen uniformly from the class. Let P mean uses Python and J mean uses Java. Find $P(P \mid J)$ and $P(J \mid P)$.
Worked solution. If 24 students use Python, 18 use Java, and 10 use both, then

$$P(P \mid J) = \frac{10}{18}, \qquad P(J \mid P) = \frac{10}{24}.$$

The shared numerator is the intersection. The event after the conditioning bar determines the denominator, so reversing the conditional usually changes the value.
Answer. $P(P \mid J) = 10/18 = 5/9$; $P(J \mid P) = 10/24 = 5/12$.

Frequent D errors

Do not add probabilities for a non-disjoint "or" without subtracting the overlap. Do not keep a denominator unchanged after sampling without replacement. Do not confuse $P(A \mid B)$ with $P(B \mid A)$ or infer independence from informal wording.

D readiness

Can I name the sample space, choose the correct denominator, use a complement or overlap identity, and distinguish conditional probability from independence?

E. Modeling, Bayes, Expectation, and Interpretation

Total probability and Bayes' rule

If $B_1, \ldots, B_m$ partition the sample space, then

$$P(A) = \sum_i P(A \mid B_i)P(B_i).$$

Bayes' rule reverses the condition:

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{P(A)}.$$

The prior or base rate $P(B_j)$ remains part of the numerator.

E.1: numerical independence test

Problem. Events A, B satisfy $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$. Determine whether they are independent, showing the numerical test, and then find $P(A \mid B)$.
Worked solution. If $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, then $P(A)P(B) = 0.3 = P(A \cap B)$, so the events are independent. Consequently $P(A \mid B) = 0.3/0.5 = 0.6 = P(A)$.
Answer. They are independent, and $P(A \mid B) = 0.6$.

E.2: screening and base rates

Problem. A condition occurs in 1% of a population. A test is positive for 95% of people who have the condition and for 8% of people who do not. For a person chosen at random from this population, find $P(\text{positive})$ and $P(\text{condition} \mid \text{positive})$. Give a one-sentence base-rate interpretation.

Worked solution. Let C be the condition and $+$ a positive test. Then

$$\underbrace{0.95(0.01)}_{\text{true positives}} = 0.0095, \qquad \underbrace{0.08(0.99)}_{\text{false positives}} = 0.0792.$$

Add the two disjoint routes: $P(+) = 0.0095 + 0.0792 = 0.0887$. Then Bayes' rule keeps the true-positive route in the numerator:

$$P(C \mid +) = \frac{0.0095}{0.0887} \approx 0.107.$$

Even a sensitive test can yield many false positives when the condition is rare; the denominator includes both true and false positives.

Answer. $P(+) = 0.0887$ and $P(C \mid +) = 95/887 \approx 10.7\%$. Most positive results are false positives because the condition is rare.

E.3: factory Bayes model

Problem. Factory A supplies 40% of a store's batteries and 3% of its batteries are defective. Factory B supplies the rest and 1% of its batteries are defective. A battery is selected uniformly from the store's stock and is found defective. Find the probability it came from A, showing total probability and Bayes' rule.

Worked solution. Factory A supplies 40% with defect rate 3%; B supplies 60% with rate 1%. First compute both joint routes:

$$P(A \cap D) = 0.40(0.03) = 0.012,$$
$$P(B \cap D) = 0.60(0.01) = 0.006.$$

Thus $P(D) = 0.018$ and $P(A \mid D) = 0.012/0.018 = 2/3$.

Answer. $P(A \mid D) = 2/3$.

Expected value

For numerical outcomes x_i with probabilities p_i,

$$E[X] = \sum_i x_i p_i.$$

Use *net* outcomes when a cost is paid. Expected value is a long-run average, not a guaranteed result from one trial.

E.4: expected net value

Problem. A game uses a fair six-sided die. It pays \$10 for a 6, \$2 for a 4 or 5, and \$0 otherwise. A play costs \$2 regardless of the result. Find the expected net value of one play and interpret its sign.

Worked solution. A fair die pays 10 on a 6, 2 on a 4 or 5, and 0 otherwise. With a 2 cost, first convert payouts to net outcomes: $8, 0, -2$, with probabilities $1/6, 2/6, 3/6$. Therefore

$$E[X] = 8\left(\frac{1}{6}\right) + 0\left(\frac{2}{6}\right) - 2\left(\frac{3}{6}\right) = \frac{1}{3}.$$

The positive sign means an average gain of \$1/3 per play over many plays.

Answer. $E[X] = \$1/3$ per play, a positive long-run average gain.

Aggregation reversal

A treatment can have the higher success rate in every subgroup but a lower overall rate when the groups receive treatments in very different proportions. Compute each within-group rate and each pooled rate separately. Then explain which high- or low-success group dominates each pooled denominator; do not say that arithmetic is contradictory.

Twenty-minute model-selection routine

For one problem from each A–E section: name the rule or model first, write an exact unsimplified expression, then calculate or interpret. Check that each denominator matches its condition and that each counted outcome appears once.

Frequent E errors

Do not ignore base rates, reverse a conditional, or assume independence. In expected value, subtract costs from every applicable outcome. In aggregated tables, compare rates rather than raw success counts.

Whole-Core readiness

I can choose among sum, product, cases, complement, permutation, and combination; justify a probability denominator; compute conditional and Bayes probabilities; test independence; calculate net expected value; and explain an aggregation reversal. A live form contains one problem from each A–E section.